Analog Signal Conditioning and Waveform Synthesis

Objective:

Design and implement an analog signal conditioning system to extract and synthesize specific waveform components from a 1 kHz square wave using first/second order low-pass, high-pass, and band-pass filters. Both active and passive filter topologies are allowed

Input Signal Specifications

Square waveform generated by a function generator: 1 kHz,0- 5V square waveform (50% duty cycle)

Output Signals Specifications

- **Output1**: Extract the DC component from the square input waveform: DC output level 2.5V±10mV
- **Output2**: Extract the 1kHz sinusoidal signal from the square input
- **Output3**: Extract the 5kHz sinusoidal signal from the square input
- **Output 4**: Combine the 1kHz fundamental and 5kHz harmonic signal to get a 5kHz signal modulated by 1kHz signal (you can adjust signal amplitude e.g. amplitude of 5kHz signal amplitude can be 10 time 1kHz signal). Use highpass filtering to extract the 5kHz signal.
- **Output 5**: Remove the DC component from the original square waveform using a passive circuit, resulting in a zero-centered AC signal.

- Plot the frequency content of square waveform using FFT in Matlab.
- For each filtering circuit, justify the choice of filter, filter order, passive/active filter topology, cutoff freq/center freq, design equations, component values.
- Determine the transfer function and plot the theoretical frequency response using Bode plot in MATLAB
- Simulate the circuit using PSpice/Multisim/Proteus.
- Implement the circuit using bread board (Bonus Marks will be given for circuit implementation using PCB)
- Measure the voltage transfer function using function generator and oscilloscope.
- Plot the measured frequency response using MATLAB.
- Compare the theoretical, simulated and measured results by plotting them on the same graph.

- Compare the simulated and measured output waveforms.
- Write a project report. Your report may have following sections: title page, abstract, design procedure, bill of material (BOM), simulated results, measured results (you may add circuit snapshot in this section), comparison of simulated and measured results, conclusion, and references.

NOTE: This is a group project ,with three members per project. Please strictly follow the project deadlines given below

Sr No	Project Tasks	Deadlines
1	Design and simulation of analog signal conditioning and waveform synthesis circuit	21st April 2025
2	Component Procurement	28th April 2025
3	Breadboard Implementation and Project Report	12th May 2025
4	PCB Implementation (Optional for Bonus Marks)	16th May 2025